THE MODULE CONSTRUCTED FROM THE DOUBLE DUAL IS INJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the chain of identifications is displayed in full and the final exactness step is stated with its two reasons, so no step is left to the reader.

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Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The construction of the module from the double dual. Same chapter, the construction of the module attached to a module and of the sheaf of homomorphisms used in it. Supplies the object whose injectivity is asserted and its description as a sheaf of homomorphisms out of a flat module.
- The hom-tensor adjunction. Same chapter and Sheaves, the adjunction between sheaf homomorphisms and tensor products of modules. Supplies the chain of identifications rewriting the hom functor into a hom functor of a tensor product with a flat module.
- Injective objects and flat modules. Same chapter, the characterization of injective modules and the exactness of tensoring by a flat module. Supplies that the target is injective and that tensoring by a flat module is exact, which makes the rewritten functor exact.

Lemma 0.1. *Let $\mathcal{O}$ be a sheaf of rings. For every $\mathcal{O}$-module $\mathcal{F}$ the $\mathcal{O}$-module $J(\mathcal{F})$ is injective.*

Proof. We have to show that the functor $\mathrm{Hom}_{\mathcal{O}}(\mathcal{G}, J(\mathcal{F}))$ is exact. Note that

$$\begin{aligned} \mathrm{Hom}_{\mathcal{O}}(\mathcal{G}, J(\mathcal{F})) &= \mathrm{Hom}_{\mathcal{O}}(\mathcal{G}, (F(\mathcal{F}^\vee))^\vee) \\ &= \mathrm{Hom}_{\mathcal{O}}(\mathcal{G}, \mathcal{H}om(F(\mathcal{F}^\vee), \mathcal{J})) \\ &= \mathrm{Hom}(\mathcal{G} \otimes_{\mathcal{O}} F(\mathcal{F}^\vee), \mathcal{J}) \end{aligned}$$

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Thus what we want follows from the fact that $F(\mathcal{F}^\vee)$ is flat and $\mathcal{J}$ is injective. $\square$